

# **II-Probabilistic Approaches Causality**

**Foundations of Neuro-Symbolic AI**

Alex Goessmann

University of Applied Science Würzburg-Schweinfurt (THWS)

June 29, 2026

# Outline

- ▶ Bayesian network hypergraphs

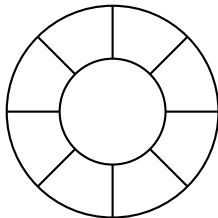
- ▶ Intervention

- ▶ Counterfactuals

## Example: Random walk

Imagine a random walk through circular cells:

- ▶  $X_k$  describes at timestep  $k \in [d]$  the position
- ▶ With probability 0.5 we stay at a cell, and to each neighboring cell with 0.25



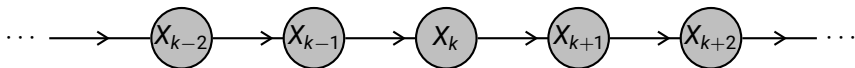
This is a Markov chain with the transition matrix:

$$\mathbb{P}[X_k | X_{k-1}] = \begin{pmatrix} 0.50 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0.25 \\ 0.25 & 0.50 & 0.25 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.25 & 0.50 & 0.25 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.25 & 0.50 & 0.25 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.25 & 0.50 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.25 & 0.50 & 0.25 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.25 & 0.50 & 0.25 \\ 0.25 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0.50 \end{pmatrix} [X_k, X_{k-1}]$$

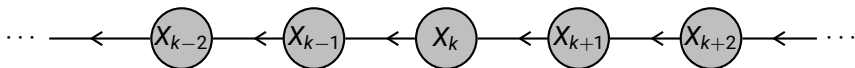
# How to distinguish the past from the present?

We have two decomposition schemes as Bayesian networks, both decorated with the same transition matrices:

Parametrization from the past to the future:



Parametrization from the future to the past:



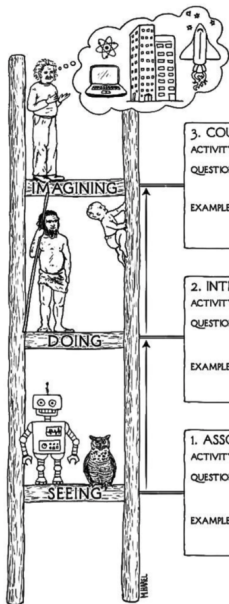
# The BN hypergraph is not unique

We can manipulate the hypergraph, and still find a Bayesian network representation of a distribution.

Causation is not (always) represented in the directionality!

**Idea:** Distinguish by interventions.

# The rungs of causation



## 3. COUNTERFACTUALS

ACTIVITY: Imagining, Retrospection, Understanding

QUESTIONS: *What if I had done ...? Why?*  
(Was it X that caused Y? What if X had not occurred? What if I had acted differently?)

EXAMPLES: Was it the aspirin that stopped my headache?  
Would Kennedy be alive if Oswald had not killed him? What if I had not smoked for the last 2 years?

→ Response variables

## 2. INTERVENTION

ACTIVITY: Doing, Intervening

QUESTIONS: *What if I do ...? How?*  
(What would Y be if I do X?  
How can I make Y happen?)

EXAMPLES: If I take aspirin, will my headache be cured?  
What if we ban cigarettes?

→ Do variables

## 1. ASSOCIATION

ACTIVITY: Seeing, Observing

QUESTIONS: *What if I see ...?*  
(How are the variables related?  
How would seeing X change my belief in Y?)

EXAMPLES: What does a symptom tell me about a disease?  
What does a survey tell us about the election results?

→ Bayesian Networks

# Bayesian networks capture associations

**Limitation:** Do not model causations.

# Outline

► Bayesian network hypergraphs

► Intervention

► Counterfactuals



# Do-calculus

Add further variables  $D_k$  representing interventions.

# Example: Random walk

**Effect in random walk:** Restart the random walk.

Random walk example with  $m = 3$ :

$$\mathbb{P}[X_k | X_{k-1}, D_k] = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0.50 & 0.25 & 0.25 \\ 0.25 & 0.50 & 0.25 \\ 0.25 & 0.25 & 0.50 \end{pmatrix} [X_k, X_{k-1}, D_k]$$

# Outline

- ▶ Bayesian network hypergraphs
- ▶ Intervention
- ▶ Counterfactuals

# Response variables

Model each conditioned probability as a basis encoding of a function on the parent variables, the do-variable and the response variable

$$L_k$$

The response variables are shared with the counterfactual world variables.

# Random walk example

We model a response variable

- ▶  $\ell_k = 0$  : going to the left cell
- ▶  $\ell_k = 0$  : staying at the current cell
- ▶  $\ell_k = 2$  : going to the right cell

$$\mathbb{P}[X_k | X_{k-1}, L_k] = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} [X_k, X_{k-1}, L_k]$$

**Effect in random walk:** Shift of the random walk.